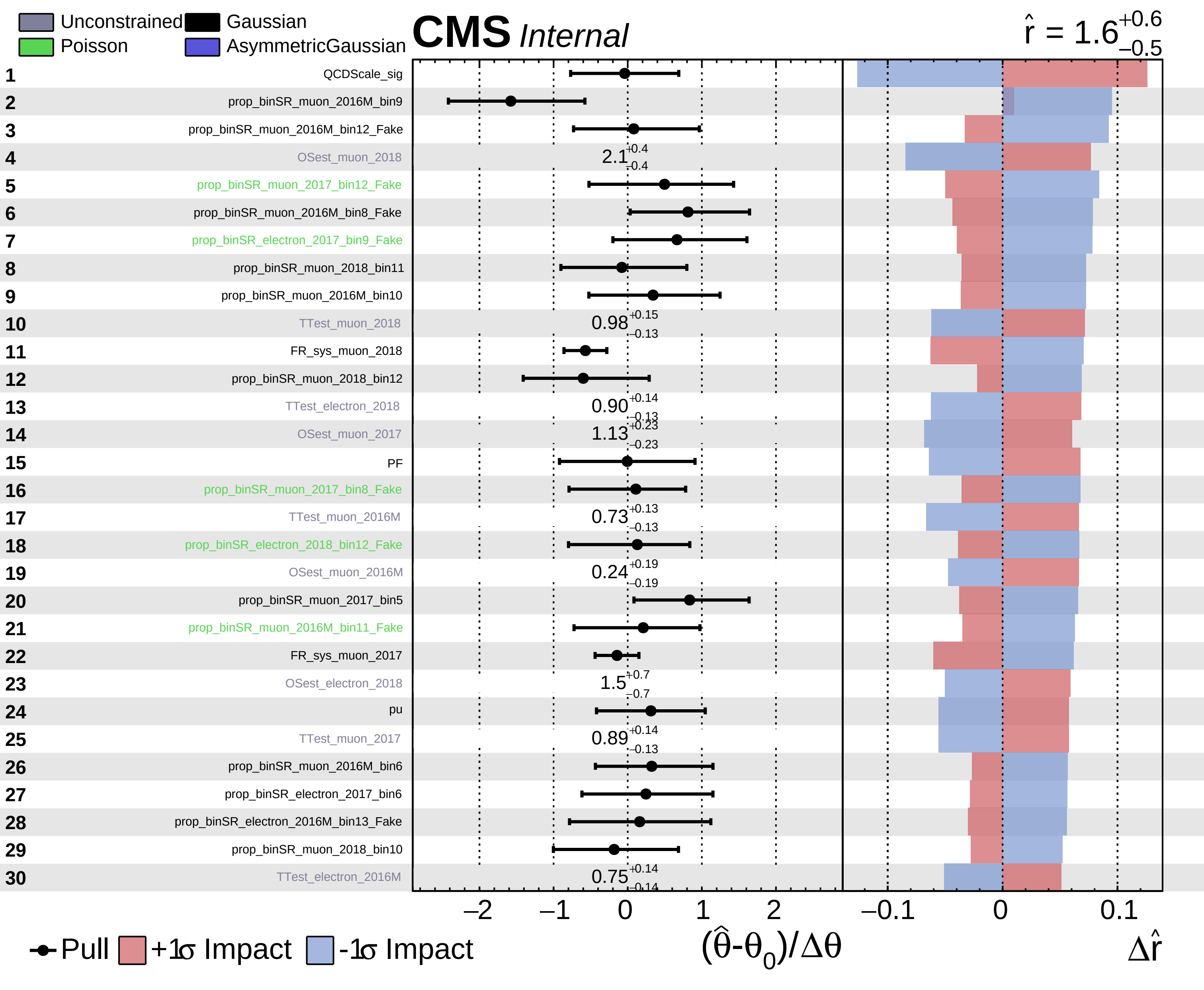
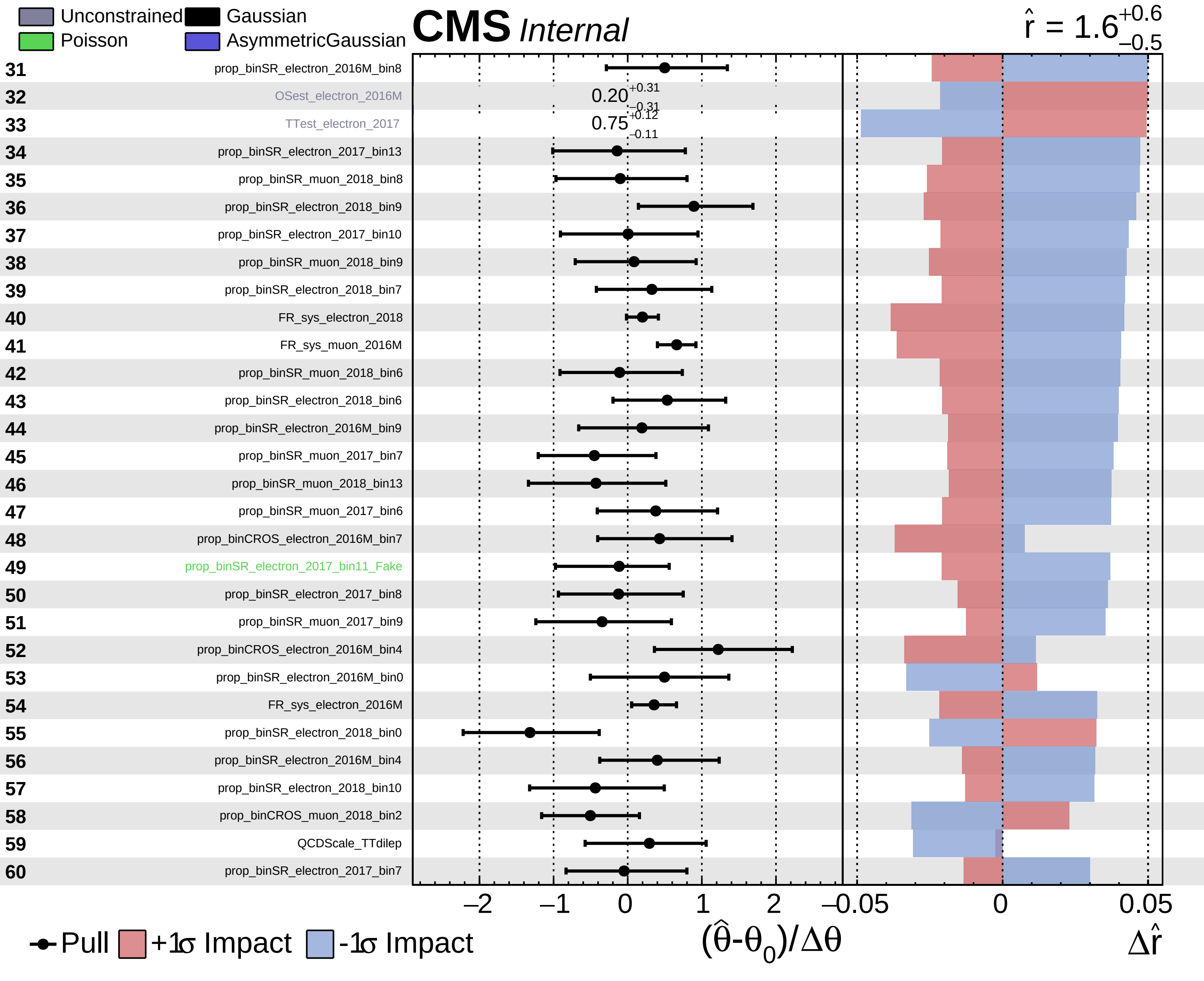
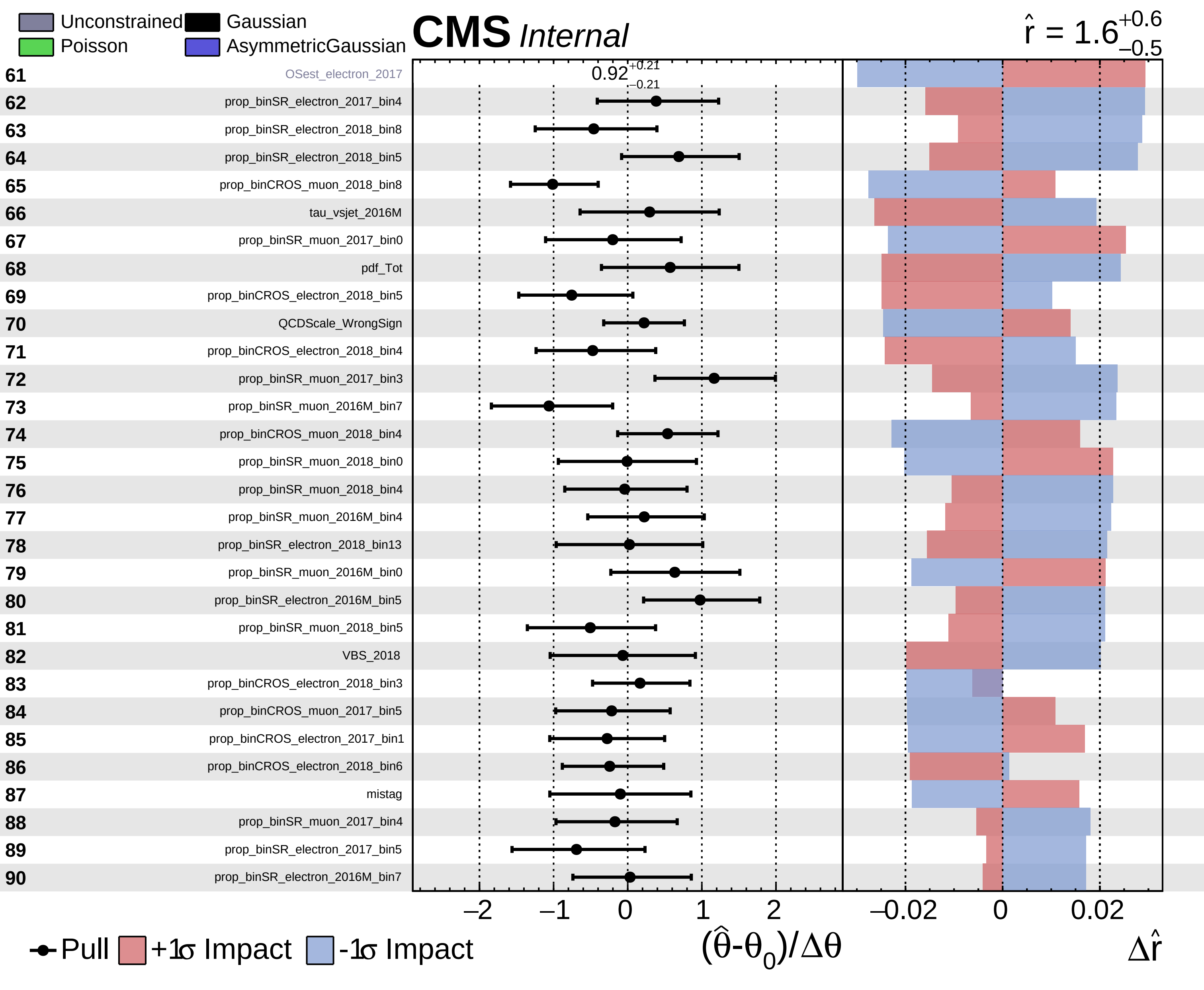


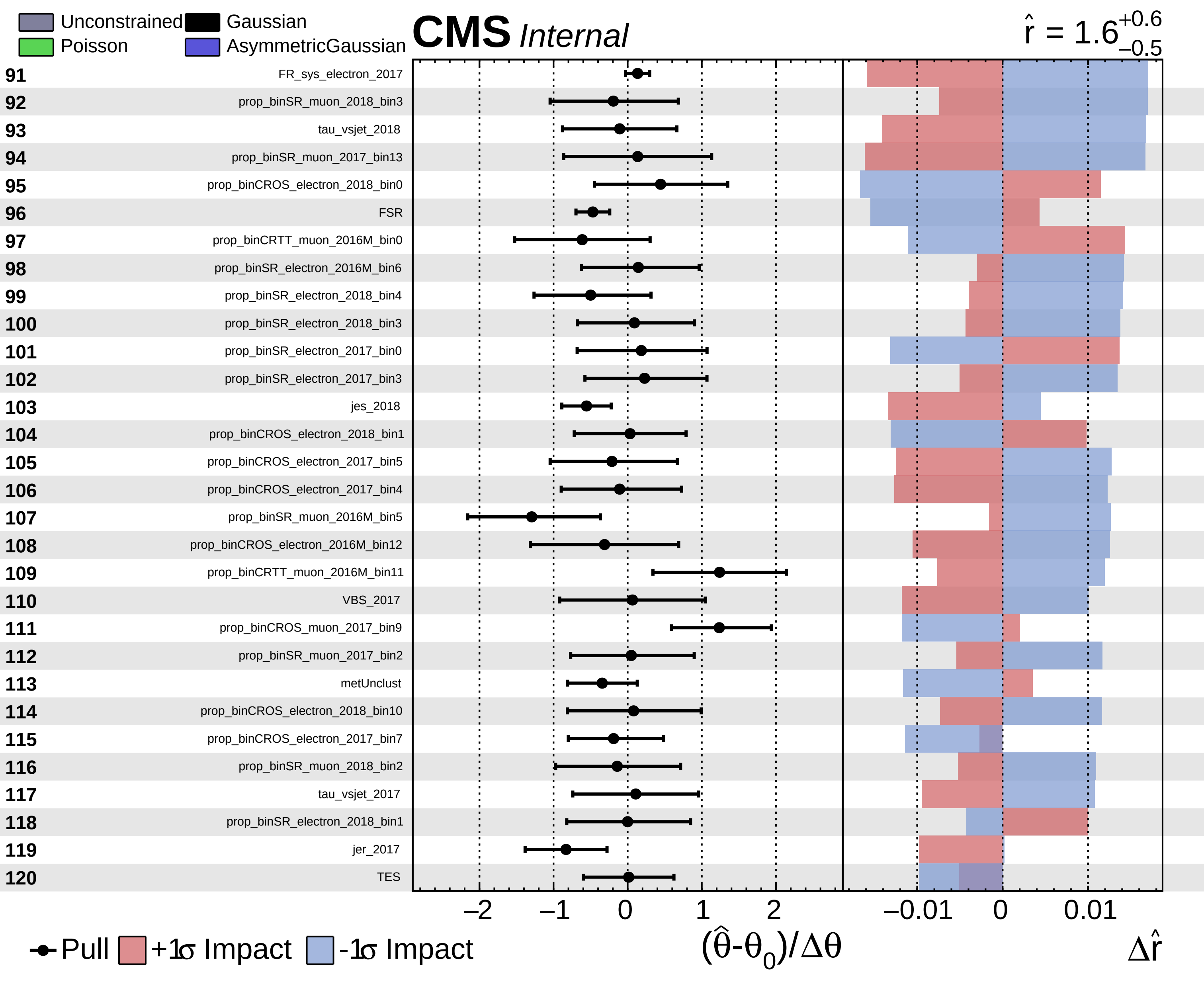
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
1 QCDScale_sig
2 prop_binSR_muon_2016M_bin9
3 prop_binSR_muon_2016M_bin12_Fake
4 OSest_muon_2018 $2.1^{+0.4}_{-0.4}$
5 prop_binSR_muon_2017_bin12_Fake
6 prop_binSR_muon_2016M_bin8_Fake
7 prop_binSR_electron_2017_bin9_Fake
8 prop_binSR_muon_2018_bin11
9 prop_binSR_muon_2016M_bin10
10 TTest_muon_2018 $0.98^{+0.15}_{-0.13}$
11 FR_sys_muon_2018
12 prop_binSR_muon_2018_bin12
13 TTest_electron_2018 $0.90^{+0.14}_{-0.13}$
14 OSest_muon_2017 $1.13^{+0.23}_{-0.23}$
15 PF
16 prop_binSR_muon_2017_bin8_Fake
17 TTest_muon_2016M $0.73^{+0.13}_{-0.13}$
18 prop_binSR_electron_2018_bin12_Fake
19 OSest_muon_2016M $0.24^{+0.19}_{-0.19}$
20 prop_binSR_muon_2017_bin5
21 prop_binSR_muon_2016M_bin11_Fake
22 FR_sys_muon_2017
23 OSest_electron_2018 $1.5^{+0.7}_{-0.7}$
24 pu
25 TTest_muon_2017 $0.89^{+0.14}_{-0.13}$
26 prop_binSR_muon_2016M_bin6
27 prop_binSR_electron_2017_bin6
28 prop_binSR_electron_2016M_bin13_Fake
29 prop_binSR_muon_2018_bin10
30 TTest_electron_2016M $0.75^{+0.14}_{-0.14}$
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



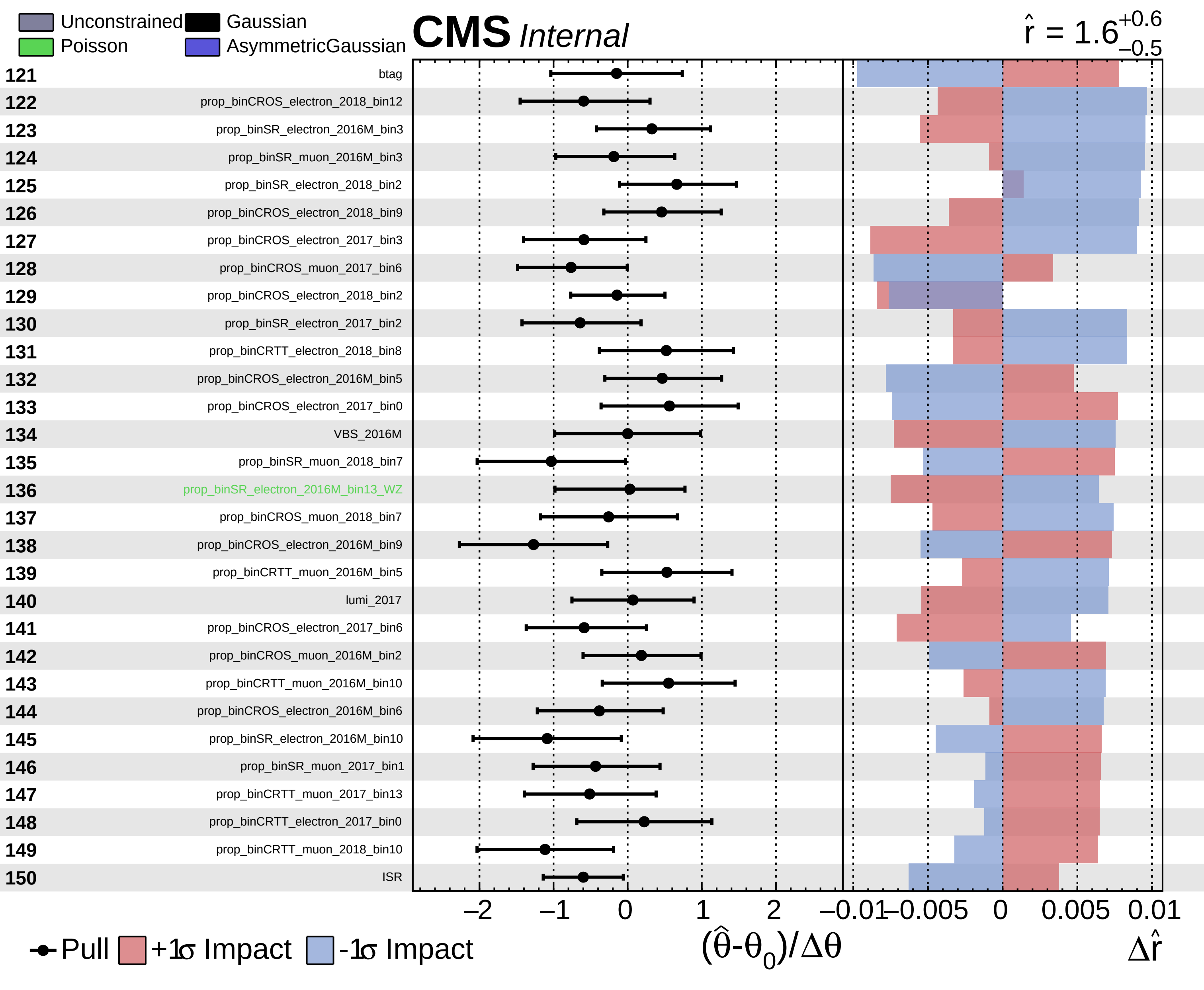
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.6 +0.6 −0.5
31 prop_binSR_electron_2016M_bin8
32 OSest_electron_2016M 0.20 +0.31 −0.31
33 TTest_electron_2017 0.75 +0.12 −0.11
34 prop_binSR_electron_2017_bin13
35 prop_binSR_muon_2018_bin8
36 prop_binSR_electron_2018_bin9
37 prop_binSR_electron_2017_bin10
38 prop_binSR_muon_2018_bin9
39 prop_binSR_electron_2018_bin7
40 FR_sys_electron_2018
41 FR_sys_muon_2016M
42 prop_binSR_muon_2018_bin6
43 prop_binSR_electron_2018_bin6
44 prop_binSR_electron_2016M_bin9
45 prop_binSR_muon_2017_bin7
46 prop_binSR_muon_2018_bin13
47 prop_binSR_muon_2017_bin6
48 prop_binCROS_electron_2016M_bin7
49 prop_binSR_electron_2017_bin11_Fake
50 prop_binSR_electron_2017_bin8
51 prop_binSR_muon_2017_bin9
52 prop_binCROS_electron_2016M_bin4
53 prop_binSR_electron_2016M_bin0
54 FR_sys_electron_2016M
55 prop_binSR_electron_2018_bin0
56 prop_binSR_electron_2016M_bin4
57 prop_binSR_electron_2018_bin10
58 prop_binCROS_muon_2018_bin2
59 QCDScale_TTdilep
60 prop_binSR_electron_2017_bin7
−2 −1 0 1 2
−0.05 0 0.05
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



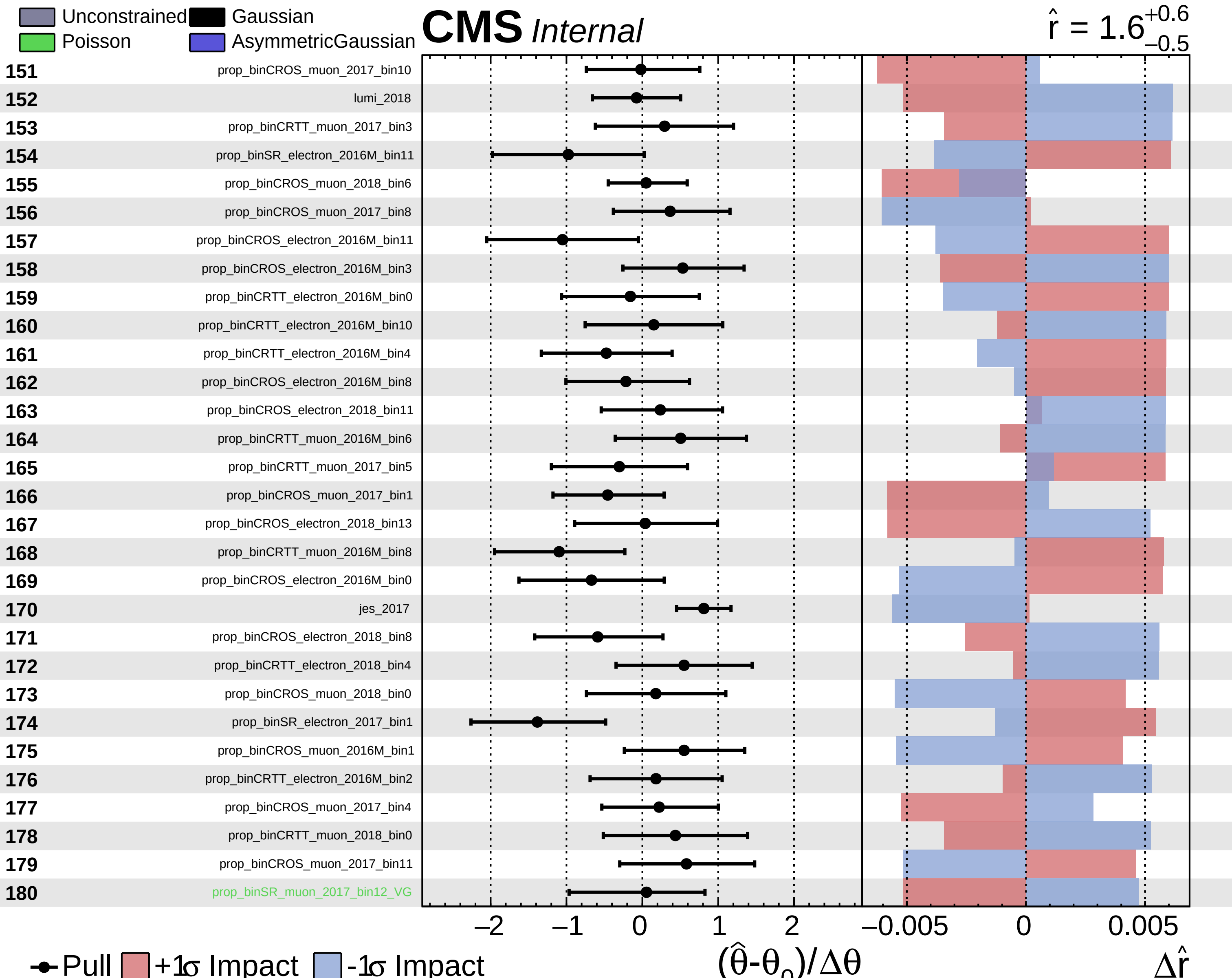
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
61 OSest_electron_2017
$0.92^{+0.21}_{-0.21}$
62 prop_binSR_electron_2017_bin4
63 prop_binSR_electron_2018_bin8
64 prop_binSR_electron_2018_bin5
65 prop_binCROS_muon_2018_bin8
66 tau_vsjet_2016M
67 prop_binSR_muon_2017_bin0
68 pdf_Tot
69 prop_binCROS_electron_2018_bin5
70 QCDScale_WrongSign
71 prop_binCROS_electron_2018_bin4
72 prop_binSR_muon_2017_bin3
73 prop_binSR_muon_2016M_bin7
74 prop_binCROS_muon_2018_bin4
75 prop_binSR_muon_2018_bin0
76 prop_binSR_muon_2018_bin4
77 prop_binSR_muon_2016M_bin4
78 prop_binSR_electron_2018_bin13
79 prop_binSR_muon_2016M_bin0
80 prop_binSR_electron_2016M_bin5
81 prop_binSR_muon_2018_bin5
82 VBS_2018
83 prop_binCROS_electron_2018_bin3
84 prop_binCROS_muon_2017_bin5
85 prop_binCROS_electron_2017_bin1
86 prop_binCROS_electron_2018_bin6
87 mistag
88 prop_binSR_muon_2017_bin4
89 prop_binSR_electron_2017_bin5
90 prop_binSR_electron_2016M_bin7
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



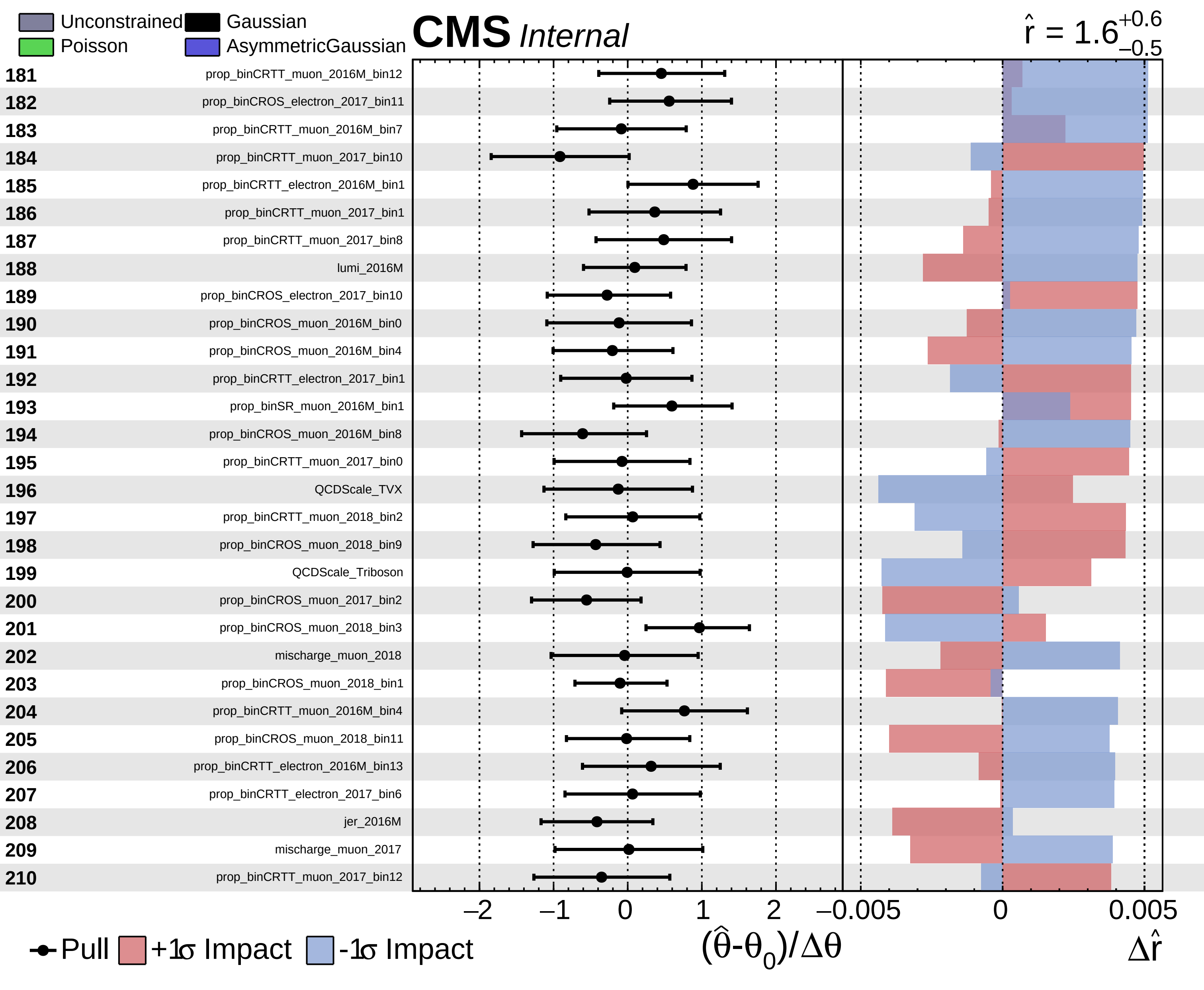
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
91 FR_sys_electron_2017
92 prop_binSR_muon_2018_bin3
93 tau_vsjet_2018
94 prop_binSR_muon_2017_bin13
95 prop_binCROS_electron_2018_bin0
96 FSR
97 prop_binCRTT_muon_2016M_bin0
98 prop_binSR_electron_2016M_bin6
99 prop_binSR_electron_2018_bin4
100 prop_binSR_electron_2018_bin3
101 prop_binSR_electron_2017_bin0
102 prop_binSR_electron_2017_bin3
103 jes_2018
104 prop_binCROS_electron_2018_bin1
105 prop_binCROS_electron_2017_bin5
106 prop_binCROS_electron_2017_bin4
107 prop_binSR_muon_2016M_bin5
108 prop_binCROS_electron_2016M_bin12
109 prop_binCRTT_muon_2016M_bin11
110 VBS_2017
111 prop_binCROS_muon_2017_bin9
112 prop_binSR_muon_2017_bin2
113 metUnclust
114 prop_binCROS_electron_2018_bin10
115 prop_binCROS_electron_2017_bin7
116 prop_binSR_muon_2018_bin2
117 tau_vsjet_2017
118 prop_binSR_electron_2018_bin1
119 jer_2017
120 TES
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

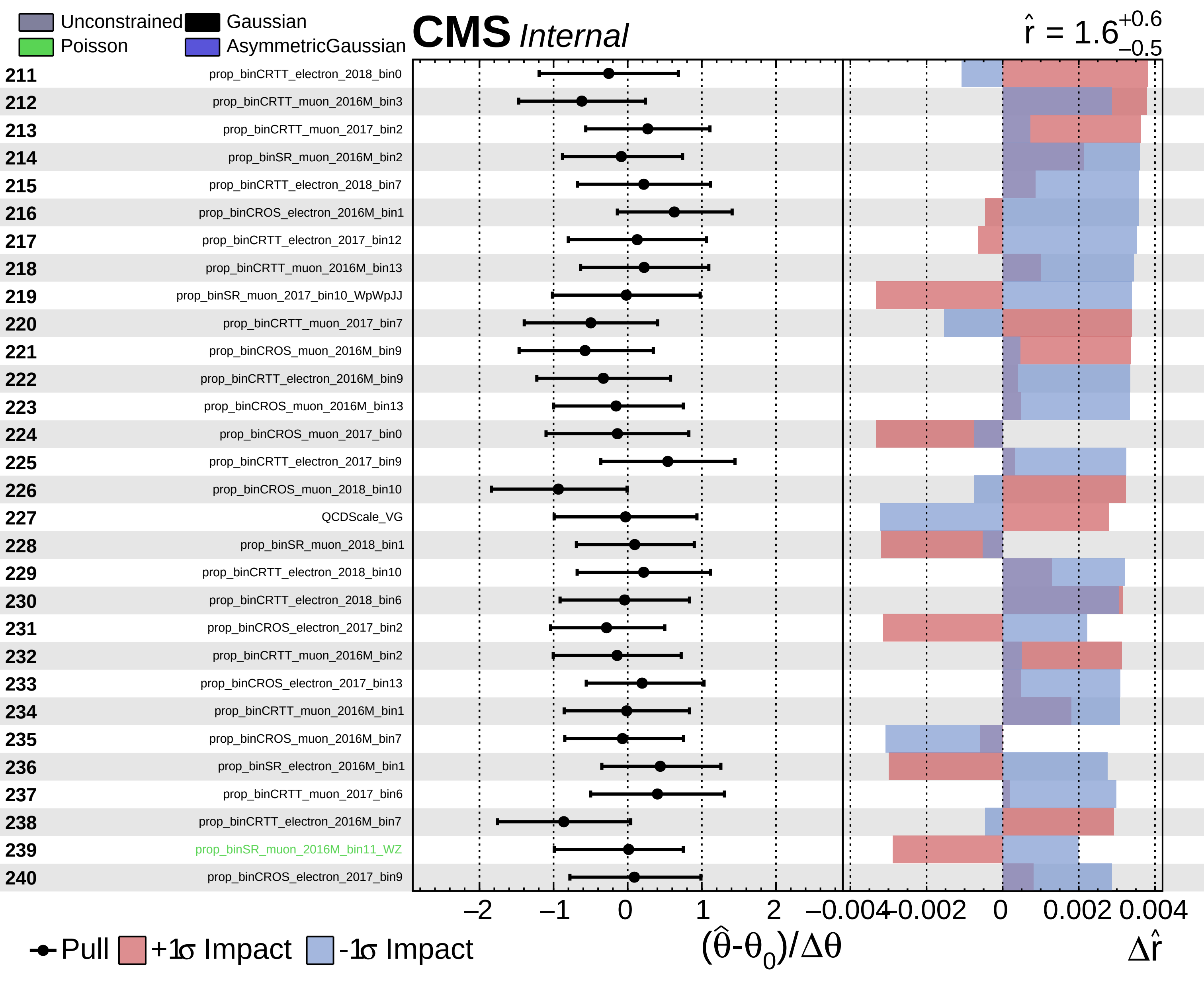


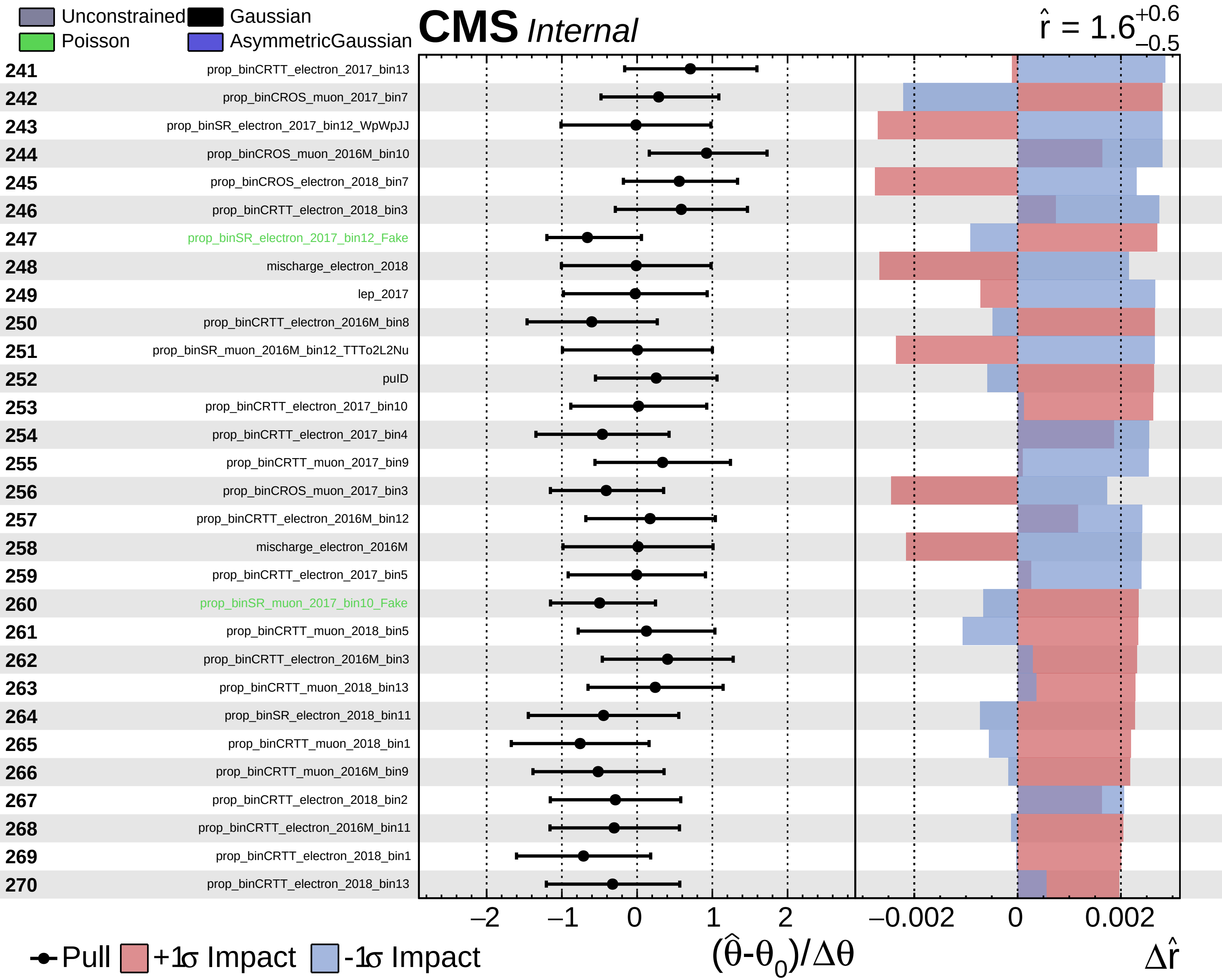
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
121 btag
122 prop_binCROS_electron_2018_bin12
123 prop_binSR_electron_2016M_bin3
124 prop_binSR_muon_2016M_bin3
125 prop_binSR_electron_2018_bin2
126 prop_binCROS_electron_2018_bin9
127 prop_binCROS_electron_2017_bin3
128 prop_binCROS_muon_2017_bin6
129 prop_binCROS_electron_2018_bin2
130 prop_binSR_electron_2017_bin2
131 prop_binCRTT_electron_2018_bin8
132 prop_binCROS_electron_2016M_bin5
133 prop_binCROS_electron_2017_bin0
134 VBS_2016M
135 prop_binSR_muon_2018_bin7
136 prop_binSR_electron_2016M_bin13_WZ
137 prop_binCROS_muon_2018_bin7
138 prop_binCROS_electron_2016M_bin9
139 prop_binCRTT_muon_2016M_bin5
140 lumi_2017
141 prop_binCROS_electron_2017_bin6
142 prop_binCROS_muon_2016M_bin2
143 prop_binCRTT_muon_2016M_bin10
144 prop_binCROS_electron_2016M_bin6
145 prop_binSR_electron_2016M_bin10
146 prop_binSR_muon_2017_bin1
147 prop_binCRTT_muon_2017_bin13
148 prop_binCRTT_electron_2017_bin0
149 prop_binCRTT_muon_2018_bin10
150 ISR
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



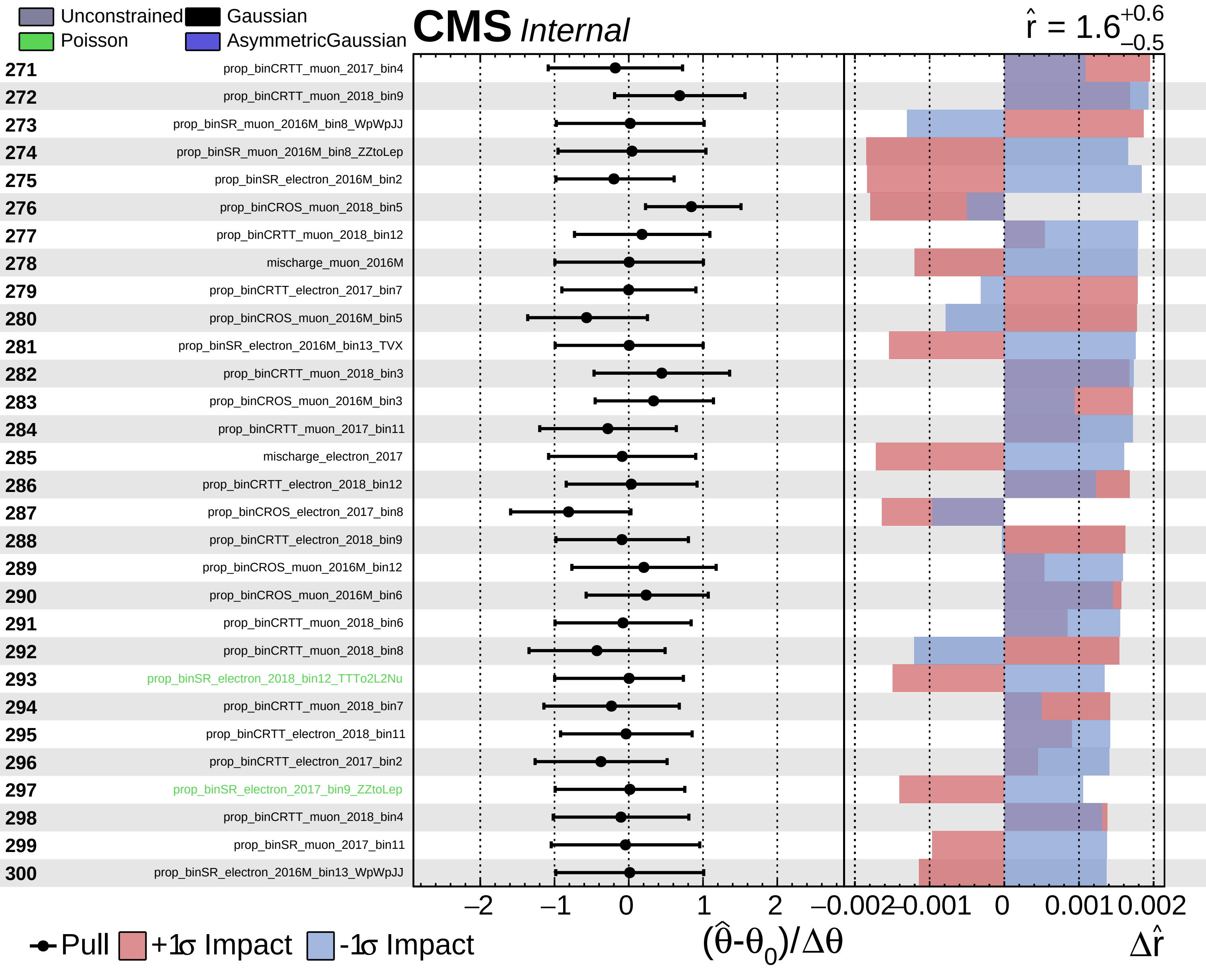
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
151 prop_binCROS_muon_2017_bin10
152 lumi_2018
153 prop_binCRTT_muon_2017_bin3
154 prop_binSR_electron_2016M_bin11
155 prop_binCROS_muon_2018_bin6
156 prop_binCROS_muon_2017_bin8
157 prop_binCROS_electron_2016M_bin11
158 prop_binCROS_electron_2016M_bin3
159 prop_binCRTT_electron_2016M_bin0
160 prop_binCRTT_electron_2016M_bin10
161 prop_binCRTT_electron_2016M_bin4
162 prop_binCROS_electron_2016M_bin8
163 prop_binCROS_electron_2018_bin11
164 prop_binCRTT_muon_2016M_bin6
165 prop_binCRTT_muon_2017_bin5
166 prop_binCROS_muon_2017_bin1
167 prop_binCROS_electron_2018_bin13
168 prop_binCRTT_muon_2016M_bin8
169 prop_binCROS_electron_2016M_bin0
170 jes_2017
171 prop_binCROS_electron_2018_bin8
172 prop_binCRTT_electron_2018_bin4
173 prop_binCROS_muon_2018_bin0
174 prop_binSR_electron_2017_bin1
175 prop_binCROS_muon_2016M_bin1
176 prop_binCRTT_electron_2016M_bin2
177 prop_binCROS_muon_2017_bin4
178 prop_binCRTT_muon_2018_bin0
179 prop_binCROS_muon_2017_bin11
180 prop_binSR_muon_2017_bin12_VG
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
181 prop_binCRTT_muon_2016M_bin12
182 prop_binCROS_electron_2017_bin11
183 prop_binCRTT_muon_2016M_bin7
184 prop_binCRTT_muon_2017_bin10
185 prop_binCRTT_electron_2016M_bin1
186 prop_binCRTT_muon_2017_bin1
187 prop_binCRTT_muon_2017_bin8
188 lumi_2016M
189 prop_binCROS_electron_2017_bin10
190 prop_binCROS_muon_2016M_bin0
191 prop_binCROS_muon_2016M_bin4
192 prop_binCRTT_electron_2017_bin1
193 prop_binSR_muon_2016M_bin1
194 prop_binCROS_muon_2016M_bin8
195 prop_binCRTT_muon_2017_bin0
196 QCDScale_TVX
197 prop_binCRTT_muon_2018_bin2
198 prop_binCROS_muon_2018_bin9
199 QCDScale_Triboson
200 prop_binCROS_muon_2017_bin2
201 prop_binCROS_muon_2018_bin3
202 mischarge_muon_2018
203 prop_binCROS_muon_2018_bin1
204 prop_binCRTT_muon_2016M_bin4
205 prop_binCROS_muon_2018_bin11
206 prop_binCRTT_electron_2016M_bin13
207 prop_binCRTT_electron_2017_bin6
208 jer_2016M
209 mischarge_muon_2017
210 prop_binCRTT_muon_2017_bin12
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

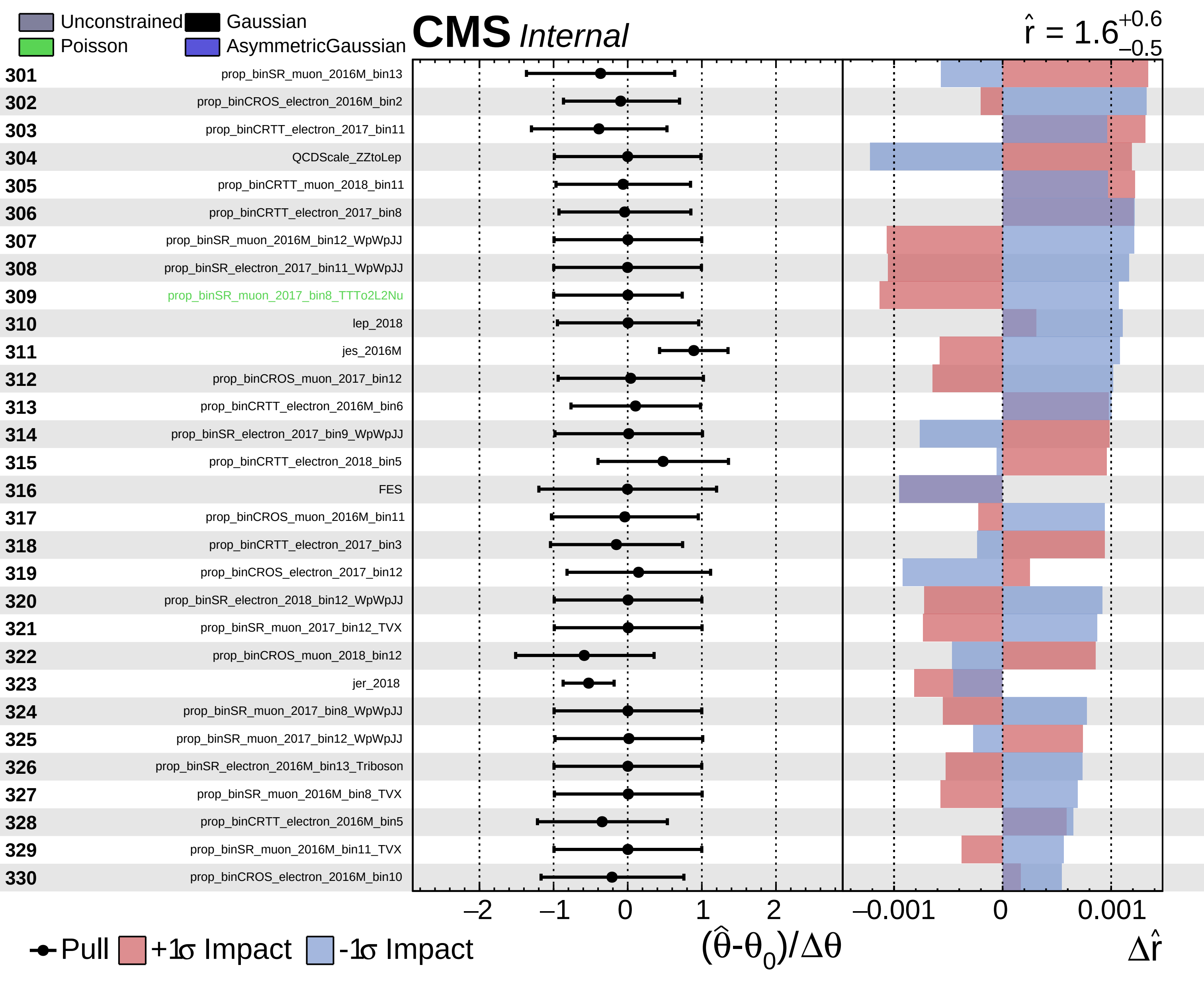

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
211 prop_binCRTT_electron_2018_bin0
212 prop_binCRTT_muon_2016M_bin3
213 prop_binCRTT_muon_2017_bin2
214 prop_binSR_muon_2016M_bin2
215 prop_binCRTT_electron_2018_bin7
216 prop_binCROS_electron_2016M_bin1
217 prop_binCRTT_electron_2017_bin12
218 prop_binCRTT_muon_2016M_bin13
219 prop_binSR_muon_2017_bin10_WpWpJJ
220 prop_binCRTT_muon_2017_bin7
221 prop_binCROS_muon_2016M_bin9
222 prop_binCRTT_electron_2016M_bin9
223 prop_binCROS_muon_2016M_bin13
224 prop_binCROS_muon_2017_bin0
225 prop_binCRTT_electron_2017_bin9
226 prop_binCROS_muon_2018_bin10
227 QCDScale_VG
228 prop_binSR_muon_2018_bin1
229 prop_binCRTT_electron_2018_bin10
230 prop_binCRTT_electron_2018_bin6
231 prop_binCROS_electron_2017_bin2
232 prop_binCRTT_muon_2016M_bin2
233 prop_binCROS_electron_2017_bin13
234 prop_binCRTT_muon_2016M_bin1
235 prop_binCROS_muon_2016M_bin7
236 prop_binSR_electron_2016M_bin1
237 prop_binCRTT_muon_2017_bin6
238 prop_binCRTT_electron_2016M_bin7
239 prop_binSR_muon_2016M_bin11_WZ
240 prop_binCROS_electron_2017_bin9
−2
−1
0
1
2
−0.004
−0.002
0
0.002
0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



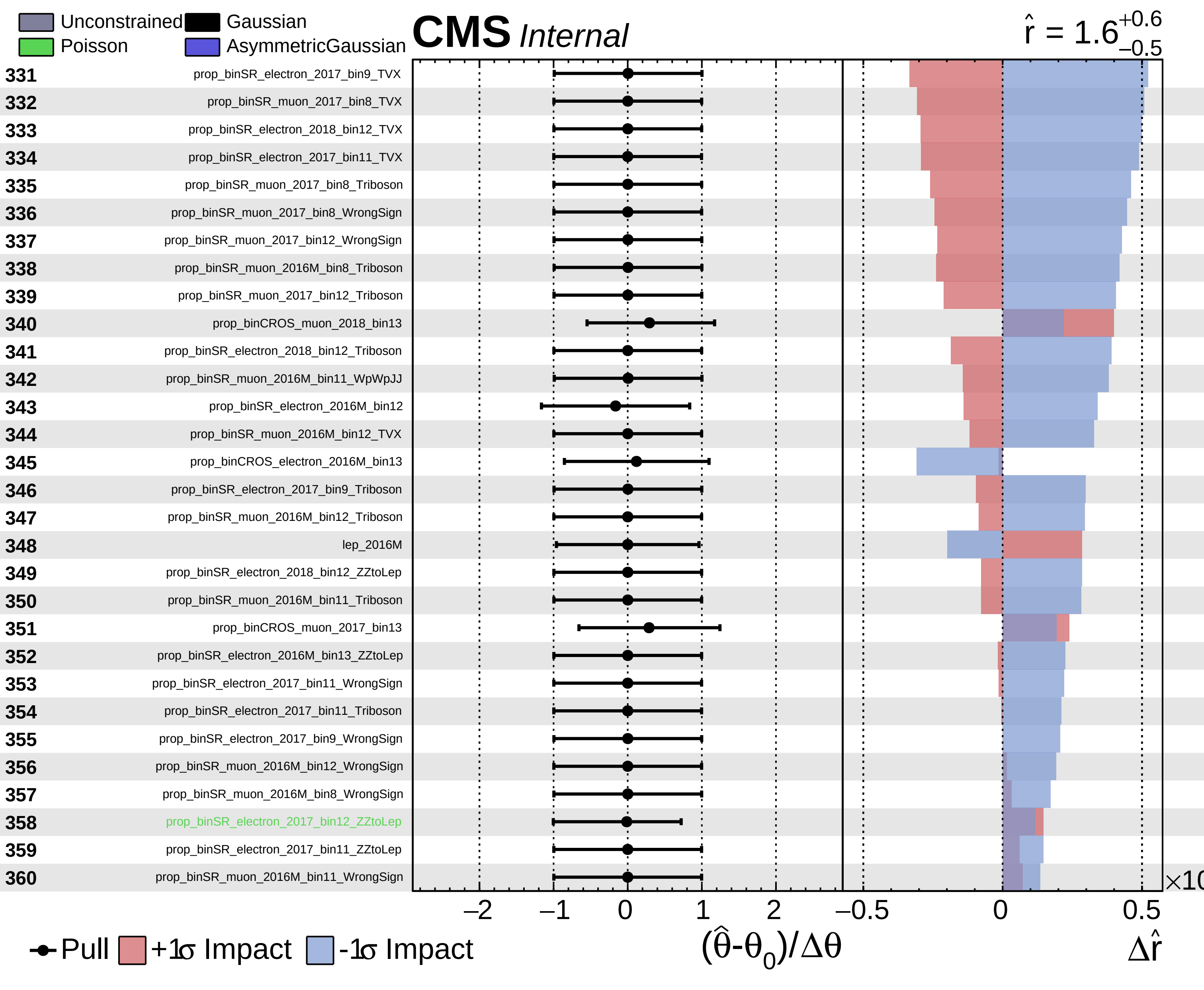
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
241 prop_binCRTT_electron_2017_bin13
242 prop_binCROS_muon_2017_bin7
243 prop_binSR_electron_2017_bin12_WpWpJJ
244 prop_binCROS_muon_2016M_bin10
245 prop_binCROS_electron_2018_bin7
246 prop_binCRTT_electron_2018_bin3
247 prop_binSR_electron_2017_bin12_Fake
248 mischarge_electron_2018
249 lep_2017
250 prop_binCRTT_electron_2016M_bin8
251 prop_binSR_muon_2016M_bin12_TTTo2L2Nu
252 puID
253 prop_binCRTT_electron_2017_bin10
254 prop_binCRTT_electron_2017_bin4
255 prop_binCRTT_muon_2017_bin9
256 prop_binCROS_muon_2017_bin3
257 prop_binCRTT_electron_2016M_bin12
258 mischarge_electron_2016M
259 prop_binCRTT_electron_2017_bin5
260 prop_binSR_muon_2017_bin10_Fake
261 prop_binCRTT_muon_2018_bin5
262 prop_binCRTT_electron_2016M_bin3
263 prop_binCRTT_muon_2018_bin13
264 prop_binSR_electron_2018_bin11
265 prop_binCRTT_muon_2018_bin1
266 prop_binCRTT_muon_2016M_bin9
267 prop_binCRTT_electron_2018_bin2
268 prop_binCRTT_electron_2016M_bin11
269 prop_binCRTT_electron_2018_bin1
270 prop_binCRTT_electron_2018_bin13
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



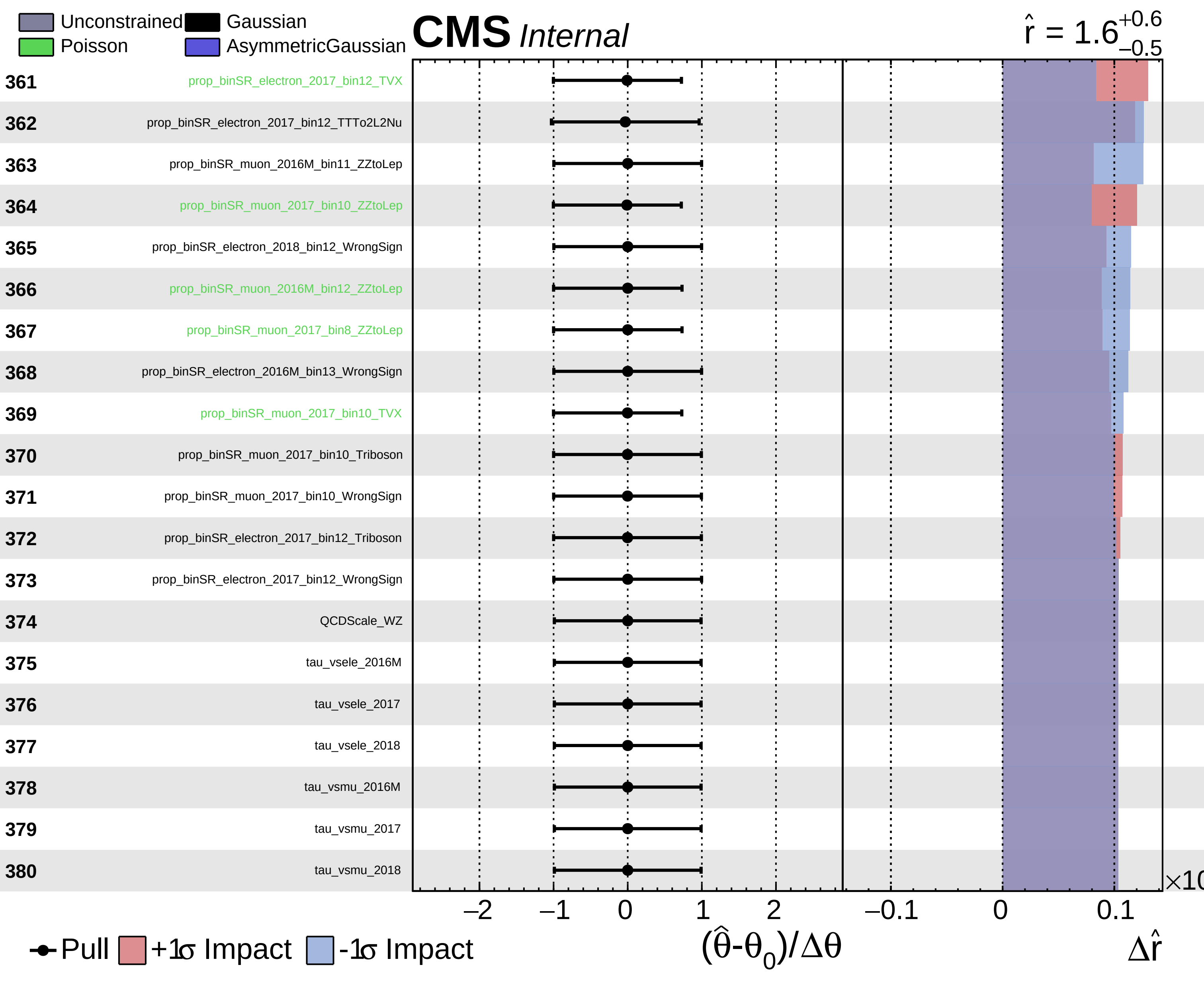
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
271 prop_binCRTT_muon_2017_bin4
272 prop_binCRTT_muon_2018_bin9
273 prop_binSR_muon_2016M_bin8_WpWpJJ
274 prop_binSR_muon_2016M_bin8_ZZtoLep
275 prop_binSR_electron_2016M_bin2
276 prop_binCROS_muon_2018_bin5
277 prop_binCRTT_muon_2018_bin12
278 mischarge_muon_2016M
279 prop_binCRTT_electron_2017_bin7
280 prop_binCROS_muon_2016M_bin5
281 prop_binSR_electron_2016M_bin13_TVX
282 prop_binCRTT_muon_2018_bin3
283 prop_binCROS_muon_2016M_bin3
284 prop_binCRTT_muon_2017_bin11
285 mischarge_electron_2017
286 prop_binCRTT_electron_2018_bin12
287 prop_binCROS_electron_2017_bin8
288 prop_binCRTT_electron_2018_bin9
289 prop_binCROS_muon_2016M_bin12
290 prop_binCROS_muon_2016M_bin6
291 prop_binCRTT_muon_2018_bin6
292 prop_binCRTT_muon_2018_bin8
293 prop_binSR_electron_2018_bin12_TTTo2L2Nu
294 prop_binCRTT_muon_2018_bin7
295 prop_binCRTT_electron_2018_bin11
296 prop_binCRTT_electron_2017_bin2
297 prop_binSR_electron_2017_bin9_ZZtoLep
298 prop_binCRTT_muon_2018_bin4
299 prop_binSR_muon_2017_bin11
300 prop_binSR_electron_2016M_bin13_WpWpJJ
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
301 prop_binSR_muon_2016M_bin13
302 prop_binCROS_electron_2016M_bin2
303 prop_binCRTT_electron_2017_bin11
304 QCDScale_ZZtoLep
305 prop_binCRTT_muon_2018_bin11
306 prop_binCRTT_electron_2017_bin8
307 prop_binSR_muon_2016M_bin12_WpWpJJ
308 prop_binSR_electron_2017_bin11_WpWpJJ
309 prop_binSR_muon_2017_bin8_TTTo2L2Nu
310 lep_2018
311 jes_2016M
312 prop_binCROS_muon_2017_bin12
313 prop_binCRTT_electron_2016M_bin6
314 prop_binSR_electron_2017_bin9_WpWpJJ
315 prop_binCRTT_electron_2018_bin5
316 FES
317 prop_binCROS_muon_2016M_bin11
318 prop_binCRTT_electron_2017_bin3
319 prop_binCROS_electron_2017_bin12
320 prop_binSR_electron_2018_bin12_WpWpJJ
321 prop_binSR_muon_2017_bin12_TVX
322 prop_binCROS_muon_2018_bin12
323 jer_2018
324 prop_binSR_muon_2017_bin8_WpWpJJ
325 prop_binSR_muon_2017_bin12_WpWpJJ
326 prop_binSR_electron_2016M_bin13_Triboson
327 prop_binSR_muon_2016M_bin8_TVX
328 prop_binCRTT_electron_2016M_bin5
329 prop_binSR_muon_2016M_bin11_TVX
330 prop_binCROS_electron_2016M_bin10
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 1.6^{+0.6}_{-0.5}$
331 prop_binSR_electron_2017_bin9_TVX
332 prop_binSR_muon_2017_bin8_TVX
333 prop_binSR_electron_2018_bin12_TVX
334 prop_binSR_electron_2017_bin11_TVX
335 prop_binSR_muon_2017_bin8_Triboson
336 prop_binSR_muon_2017_bin8_WrongSign
337 prop_binSR_muon_2017_bin12_WrongSign
338 prop_binSR_muon_2016M_bin8_Triboson
339 prop_binSR_muon_2017_bin12_Triboson
340 prop_binCROS_muon_2018_bin13
341 prop_binSR_electron_2018_bin12_Triboson
342 prop_binSR_muon_2016M_bin11_WpWpJJ
343 prop_binSR_electron_2016M_bin12
344 prop_binSR_muon_2016M_bin12_TVX
345 prop_binCROS_electron_2016M_bin13
346 prop_binSR_electron_2017_bin9_Triboson
347 prop_binSR_muon_2016M_bin12_Triboson
348 lep_2016M
349 prop_binSR_electron_2018_bin12_ZZtoLep
350 prop_binSR_muon_2016M_bin11_Triboson
351 prop_binCROS_muon_2017_bin13
352 prop_binSR_electron_2016M_bin13_ZZtoLep
353 prop_binSR_electron_2017_bin11_WrongSign
354 prop_binSR_electron_2017_bin11_Triboson
355 prop_binSR_electron_2017_bin9_WrongSign
356 prop_binSR_muon_2016M_bin12_WrongSign
357 prop_binSR_muon_2016M_bin8_WrongSign
358 prop_binSR_electron_2017_bin12_ZZtoLep
359 prop_binSR_electron_2017_bin11_ZZtoLep
360 prop_binSR_muon_2016M_bin11_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 1.6 +0.6 −0.5
361 prop_binSR_electron_2017_bin12_TVX
362 prop_binSR_electron_2017_bin12_TTTo2L2Nu
363 prop_binSR_muon_2016M_bin11_ZZtoLep
364 prop_binSR_muon_2017_bin10_ZZtoLep
365 prop_binSR_electron_2018_bin12_WrongSign
366 prop_binSR_muon_2016M_bin12_ZZtoLep
367 prop_binSR_muon_2017_bin8_ZZtoLep
368 prop_binSR_electron_2016M_bin13_WrongSign
369 prop_binSR_muon_2017_bin10_TVX
370 prop_binSR_muon_2017_bin10_Triboson
371 prop_binSR_muon_2017_bin10_WrongSign
372 prop_binSR_electron_2017_bin12_Triboson
373 prop_binSR_electron_2017_bin12_WrongSign
374 QCDScale_WZ
375 tau_vsele_2016M
376 tau_vsele_2017
377 tau_vsele_2018
378 tau_vsmu_2016M
379 tau_vsmu_2017
380 tau_vsmu_2018
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂